

The Impact of Monetary Policy on Wealth Inequality

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Abstract

This paper analyzes the impact of monetary policy on wealth inequality in the United States from 1989 to 2023 using a local projection approach. We use a quarterly time series of wealth distribution from the Distributional Financial Accounts and monetary policy shocks from Bauer & Swanson (2023). Our findings indicate that expansionary monetary policy shocks exacerbate wealth inequality by disproportionately benefiting households at the top of the income distribution. However, this does not imply absolute wealth losses for lower-income households; rather, wealth accumulation occurs at a significantly faster rate among wealthier households, thereby widening inequality.

JEL: E32, D31, E52

Keywords: Wealth distribution, heterogeneous agents, monetary policy shocks

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1 Introduction

Historically, monetary policy was thought to have minimal long-term effects on wealth distribution. However, recent research challenges this view, arguing that monetary policy influences inequality through its effects on income, employment, asset prices, and interest rates (McKay & Wolf, 2023a). Concurrently, empirical evidence indicates that wealth inequality in the United States has increased significantly over the past four decades (Saez & Zucman, 2016; Batty *et al.*, 2019; Blanchet *et al.*, 2022; Smith *et al.*, 2023; Auten & Splinter, 2024; Favara *et al.*, 2025). A growing body of literature explores the factors driving this trend, with several studies identifying the interaction between interest rates and inequality as a key mechanism through which monetary policy shapes wealth distribution and economy (Gornemann *et al.*, 2016; Kaplan *et al.*, 2018). In more detail, Auclert (2019) and Bilbiie (2020) suggest that monetary policy can either dampen or amplify economic fluctuations depending on how different groups in the population are affected. Building on this research and given the central role of inequality in the HANK framework, this paper examines the impact of monetary policy on wealth inequality.

Monetary policy transmits its effects on wealth inequality through multiple channels. Ampudia *et al.* (2018), Kaplan *et al.* (2018), and Slacalek *et al.* (2020) generally categorize these into two primary mechanisms: direct and indirect. The direct channel (broadly) operates through changes in interest rates that affect how households save and on households' net financial income, assuming employment stays the same. In contrast, the indirect (general equilibrium) channel operates via employment and wage adjustments, which, in turn, affect household consumption and wealth accumulation.

Similarly, Auclert (2019) highlights how monetary policy redistributes wealth through unhedged interest rate exposure (URE), which measures the sensitivity of household balance sheets to fluctuations in real interest rates.¹ Auclert (2019) and others also show that inflation, which can result from expansionary monetary policy, shifts wealth between borrowers and lenders. When inflation rises unexpectedly, debtors gain as the real value of their debt declines, while creditors lose. These redistributive effects are inherently linked to portfolio composition, as changes in interest rates and inflation influence asset and liability classes asymmetrically. Labor market dynamics also play a crucial role in shaping wealth inequality. For a comprehensive review of these transmission mechanisms, see Colciago *et al.* (2019).

Several studies have empirically examined these effects. Coibion *et al.* (2017) found expansionary monetary policy reduces income and consumption inequality. Similarly, Piketty & Zucman (2015) argue that higher returns on capital disproportionately benefit wealthier individuals, thereby exacerbating wealth concentration. Chang & Schorfheide (2024), on the other hand, report that while expansionary policy increases consumption inequality, it reduces earnings inequality by lowering unemployment and boosting income among previously unemployed individuals. Amberg *et al.* (2022) present evidence from Sweden indicating a U-shaped

¹Gomez & Gouin-Bonenfant (2024) show theoretically that lower interest rates increase top wealth inequality when wealthier individuals borrow more than they lend. Di Maggio *et al.* (2017) emphasize the importance of household balance sheets and mortgage contract terms in shaping how monetary policy is transmitted.

effect of monetary policy on income distribution, with the highest and lowest income groups experiencing the greatest benefits, while middle-income groups receive smaller gains. Gomez & Gouin-Bonenfant (2024) employing a sufficient statistics approach, estimate a strong direct effect of lower interest rates on top wealth inequality.

To analyze the relationship between monetary policy and inequality, we employ high-frequency variations in interest rates, building on the high-frequency identification methodology for monetary policy shocks pioneered by Kuttner (2001) and subsequently refined by Gürkaynak *et al.* (2005) and others. Specifically, we utilize the dataset constructed by Bauer & Swanson (2023), which captures asset price fluctuations in the narrow time window surrounding Federal Open Market Committee (FOMC) announcements. The underlying assumption is that these short-term price movements primarily reflect unexpected shifts in Federal Reserve policy-encompassing changes in policy rates, asset purchases, and broader macroeconomic outlook revisions. If financial markets are efficient, then public information and systematic monetary policy responses should already be incorporated into asset prices, ensuring that observed movements correspond to genuine monetary policy surprises rather than extraneous noise. Bauer & Swanson (2023) further refine this approach by orthogonalizing these shocks with respect to publicly available data preceding the announcements.

For empirical assessment of wealth distribution, we rely on the Distributional Financial Accounts (DFAs) dataset (Batty *et al.*, 2019), which provides quarterly household balance sheet data segmented by wealth and income from 1989 onward. These data integrate information from the Financial Accounts of the United States and the Survey of Consumer Finances (SCF), offering high-frequency insights into wealth distribution trends than other datasets. Our analysis specifically examines the impact of monetary policy shocks on household wealth across different income groups. However, potential limitations exist, including the reliance on SCF-based interpolation and uncertainty regarding household mobility across wealth percentiles.

A key distinction in our approach is our direct focus on wealth effects rather than income or consumption effects and our reliance on state-of-the-art measures of monetary policy shocks. By leveraging high-frequency identification techniques and comprehensive household balance sheet data, our study provides new insights into the role of monetary policy in shaping wealth inequality.

2 Data

Our empirical analysis leverages a quarterly time series dataset spanning the period from 1989:Q3 to 2023:Q4, enabling us to examine wealth distribution trends in the US over approximately the last three decades.

To measure wealth distribution, we rely on the Distributional Financial Accounts (DFAs)², which integrate data from two key statistical products of the Federal Reserve Board: (i) the Financial Accounts of the United States, which provide quarterly aggregate measures of house-

²The data are available from <https://www.federalreserve.gov/releases/z1/dataviz/dfa/index.html>

hold wealth, and (ii) the Survey of Consumer Finances (SCF), which offers triennial wealth distribution estimates. The DFAs align the SCF’s wealth distribution estimates with the Financial Accounts’ balance sheet structure, ensuring conceptual consistency between the two sources. Since the SCF is conducted only every three years, the DFAs interpolate and forecast the reconciled SCF balance sheet for intervening quarters using information from the Financial Accounts and other sources. We specifically use household wealth and its components (in relative percentage shares) across different income percentiles. Specifically, we examine the wealth (net worth), assets, and liabilities of households.

To identify monetary policy shocks, we employ the updated high-frequency monetary policy surprises dataset constructed by Bauer & Swanson (2023). This dataset captures intraday fluctuations in asset prices and yields around Federal Open Market Committee (FOMC) announcements, isolating unexpected changes in policy. To address potential endogeneity concerns, Bauer & Swanson (2023) remove components of monetary policy surprises that are correlated with key economic and financial indicators, including non-farm payroll surprises, employment growth, the S&P 500 index, the yield curve slope, commodity prices, and Treasury skewness. Specifically, we utilize the intraday changes in the first four quarterly Eurodollar futures contracts (ED1–ED4) and extract the first principal component of these changes. Since we use the first principal component, our results effectively represent a weighted average of the target and path factors.³ The principal component is rescaled so that a one-unit change corresponds to a 1 percentage point change in the ED4 rate. To facilitate our analysis, we aggregate these surprises to a quarterly frequency.

Additional macroeconomic controls include the unemployment rate, real GDP, and consumer price index (CPI) data, all sourced from the Federal Reserve Economic Data (FRED) database. We use seasonally adjusted end-of-quarter CPI values (CPIAUCSL) and real GDP (GDPC1) to compute annualized quarter-on-quarter inflation and GDP growth rates, and we incorporate seasonally adjusted end-of-quarter unemployment rates in levels. Additional information regarding the data used can be found in the online Appendix.

3 Empirical Methodology

To examine the impact of monetary policy shocks on wealth distribution, we estimate impulse response functions (IRFs) using the local projection (LP) method. This approach, introduced by Jordà (2005), directly captures the dynamic effects of shocks, allowing us to estimate the average impact of monetary policy changes on wealth distribution across different income percentiles.

The LP method involves estimating a sequence of regressions for various time horizons.

³Additionally, our estimates may capture some effects of unconventional monetary policy (Swanson, 2021). However, given the lack of data beyond the one-year horizon and the fact that unconventional policies primarily influence the longer end of the yield curve, their impact on our findings is likely limited. Moreover, evidence from monetary policy shocks is typically only indirectly informative about the effects of systematic monetary policy, relying on specific assumptions (McKay & Wolf, 2023b). Thus, these shocks do not necessarily capture the broader and more persistent implications of monetary policy changes for inequality. Furthermore, other factors, such as tax policies and financial deregulation, may also contribute to wealth inequality and interact with the effects of monetary policy.

Specifically, for each horizon h and each income percentile p , we estimate the following empirical specification:

$$Y_{p,t+h} = \alpha_{p,h} + \beta_{p,h}mps_t + \gamma_h(L)W_{t-1} + \sum_{p=1}^{p=n} \Gamma_{h,p}(L)Y_{p,t-1} + e_{p,t+h} \quad (1)$$

where $Y_{p,t+h}$ denote the relative wealth/net worth share (or alternatively, asset or liability shares) of income percentile p at time $t + h$, where percentiles are indexed from 1 to n . The terms $\alpha_{p,h}$ and $e_{p,t+h}$ denote the intercept and the idiosyncratic error term, respectively. The key explanatory variable, mps_t , is the monetary policy shock at time t . The coefficients of interest $\beta_{p,h}$ quantify the direct impact of the monetary policy shock on wealth (assets/liabilities) over horizons $h = 0, 1, \dots, 12$, i.e., up to three years. Since the analysis is based on relative shares, the coefficients $\beta_{p,h}$ should theoretically sum to zero across different income percentiles. We present results for both cases: when this restriction is not imposed and when it is enforced (restricted estimation).

The term W_{t-1} , with its associated coefficients $\gamma_h(L)$, represents a set of additional control variables, including lagged values of unemployment, inflation, and GDP growth. We also include lagged endogenous variables, $Y_{p,t-1}$ for $p = 1, \dots, n$, with estimated coefficients $\Gamma_h(L)$, to account for persistence in wealth distribution dynamics.⁴ Both $\gamma_h(L)$ and $\Gamma_{h,p}(L)$ are lag polynomials, with the lag structure selected based on the AIC and BIC. We primarily employ a one-lag specification, and as a robustness check, we also estimate models with two lags. (In the online Appendix, we also provide several subsample results (excluding COVID-19, zero lower bound period). Additionally, we present a detailed decomposition of wealth changes across asset and liability classes to further examine the transmission channels of monetary policy on wealth.) We estimate this system of equations jointly (for each h) using the Generalized Method of Moments (GMM). To ensure robust inference, we report heteroskedasticity and autocorrelation-consistent (HAC) standard errors.

4 Results

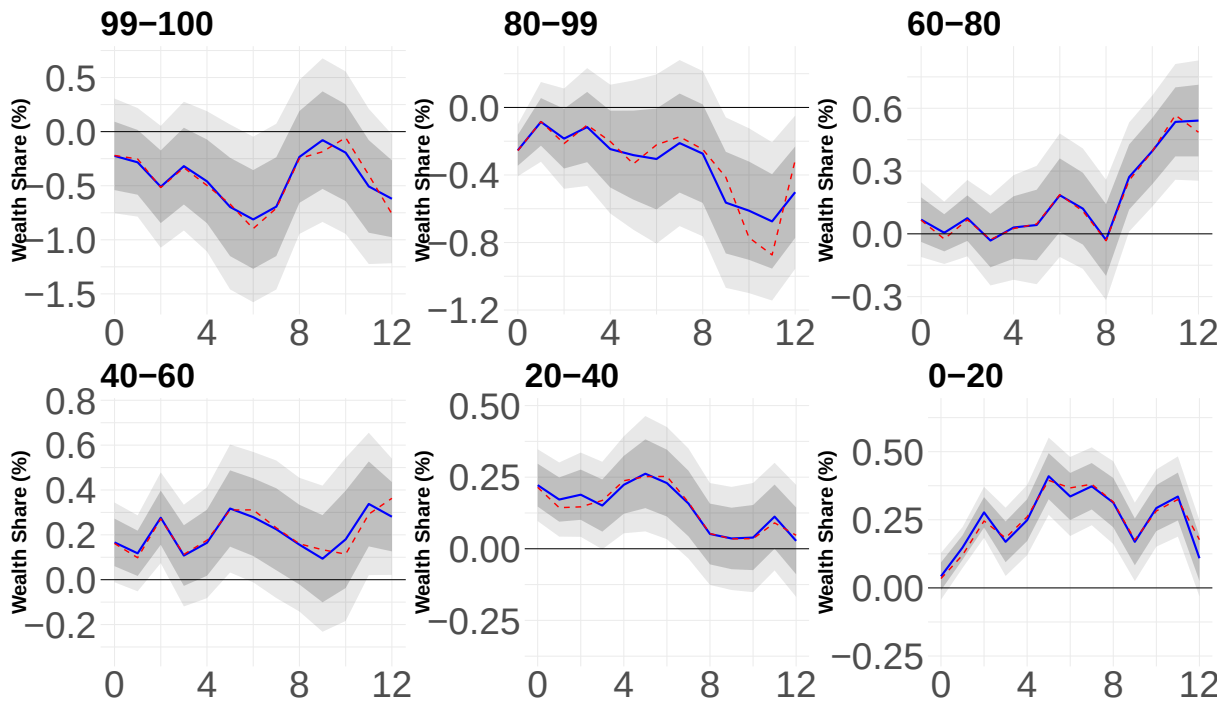
In this part, we present our empirical findings on the relationship between monetary policy and wealth inequality. We begin by analyzing the effects on wealth (net worth), followed by a disaggregated examination of household assets and liabilities to gain further insight into the distributional consequences of monetary policy.

Panel 1 presents the estimated effects of monetary policy shocks on relative wealth. The figures in the panel correspond to different income percentiles, with impulse response estimates reported for each percentile at various time horizons (indicated by the title in each figure). Results are shown for both the unrestricted (blue line) and restricted (red line) estimations. (The results are almost identical.) HAC standard errors are used to calculate confidence intervals for the unrestricted estimation. We report results for one-lag specification.

⁴Since the lagged endogenous variable appears on the right-hand side, the IRFs should be interpreted as cumulative (Jordà, 2023).

The results indicate that a 1% monetary policy shock has a mostly significant effect on wealth distribution (considering 68% confidence intervals). Specifically, our findings indicate that restrictive monetary policy shocks reduce the relative wealth of households in the 99–100 and 80–99 income percentile groups, with the most pronounced effect observed among the top 1% (99–100 income percentile). In contrast, households below the 80th income percentile generally experience relative wealth gains at the expense of these higher-income groups, with the 60–80 income percentile benefiting the least. These effects are also economically substantial; for instance, the top 1% increased its relative wealth share by 6.6% over the period under study. Additionally, it is noteworthy that the size of the effect increases over time, suggesting that the impact of shocks propagates gradually.

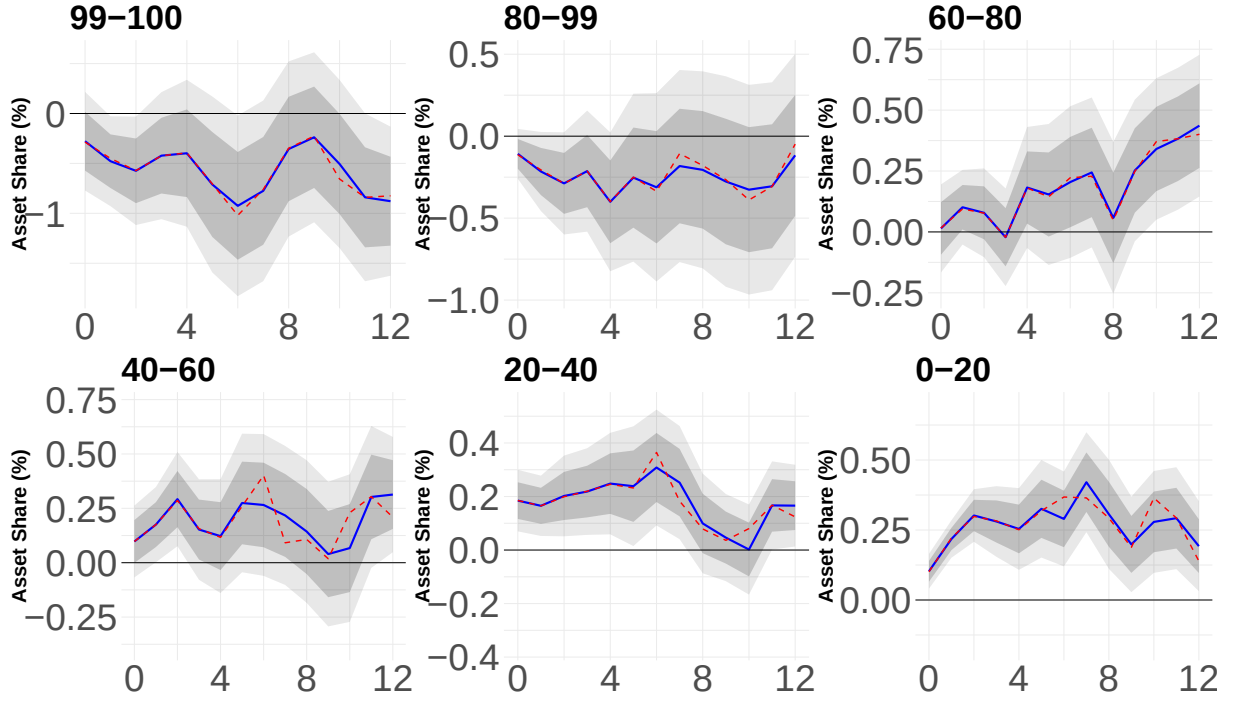
Figure 1: LP Results - Wealth (Net Worth)



Note: The figures in the panel display the results for relative wealth share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2023:Q4. The number of lags for control variables is set to 1.

Panel 2 presents the same results for assets (relative percentage shares across income percentile groups). The findings are generally consistent with those for overall wealth, and if anything, the effects are somewhat bigger. Specifically, positive monetary policy shocks reduce the relative asset shares of households in the 99–100 and 80–99 income percentile groups. The differences are likely driven by the composition of household assets across income groups. Wealthier households hold a larger share of their assets in equities and mutual funds. In contrast, lower-income households have a greater proportion of their wealth tied up in real estate and pension entitlements. This also suggests that additional adjustments on the liability side contribute to the overall impact on net worth.

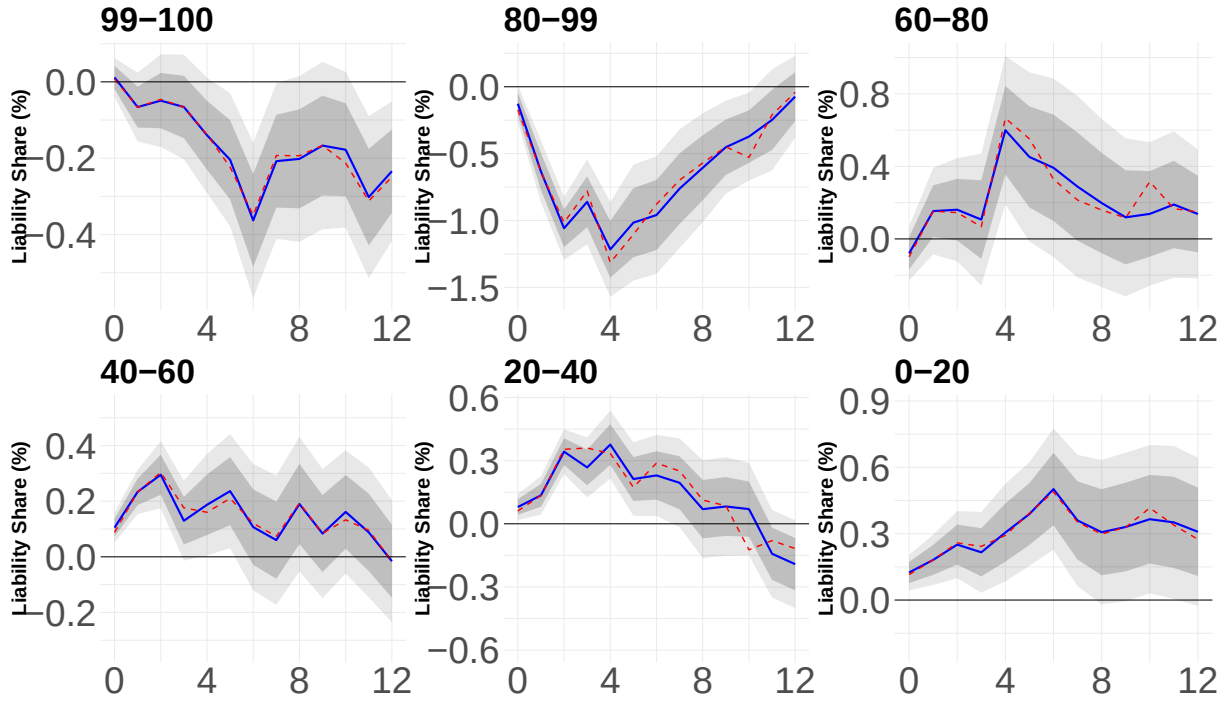
Figure 2: LP Results - Assets



Note: The figures in the panel display the results for relative asset share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2023:Q4. The number of lags for control variables is set to 1.

Finally, Panel 3 presents the results for liabilities (relative percentage shares across income percentile groups). The findings indicate that an increase in monetary policy rates reduces the liabilities of the wealthiest households, while a decrease in rates leads to higher liabilities for these groups (99–100 and 80–99 income percentiles). This result is intuitive, as a significant portion of liabilities among high-income households consists of home mortgages, which typically have fixed interest rates. Additionally, the top 1% hold a considerable share of other liabilities, suggesting that they may have greater flexibility to deleverage in response to monetary policy changes. In contrast, lower-income households rely more on consumer credit, which is more sensitive to fluctuations in interest rates. Therefore, while restrictive monetary policy reduces the liability shares of wealthy individuals, the effect on assets is even stronger, leading to a net decrease in their relative wealth. Conversely, expansionary policy has the opposite effect, as lower interest rates encourage wealthier households to accumulate assets more rapidly while taking on proportionally less debt.

Figure 3: LP Results - Liabilities



Note: The figures in the panel display the results for relative liability share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2023:Q4. The number of lags for control variables is set to 1.

Overall, our findings somewhat diverge from much of the existing literature. We find that expansionary (contractionary) monetary policy shocks contribute to increased (decreased) wealth inequality, with wealth (de)accumulation effects being more pronounced among the highest-income groups.⁵ Additionally, our results suggest a very broadly monotonically increasing effect on relative wealth across the income distribution, though some quantile crossing occurs.⁶ We interpret these findings as follows: while expansionary monetary policy may reduce earnings inequality by lowering unemployment and boosting income, the asset appreciation effect appears to dominate, leading to an overall increase in both consumption and wealth inequality, in line with Chang & Schorfheide (2024). Furthermore, Berg *et al.* (2021) theorize that older, wealthier households respond more to monetary policy shocks because their portfolios are tilted toward long-term assets (real estate, pension entitlements, and equities), which are more sensitive to interest rate changes. In contrast, younger households primarily finance consumption through labor income. These findings also seem to coincide with our result.

⁵Note that the linearity of our model specification implies that an unanticipated monetary policy easing has a symmetric and opposite effect on the inequality compared to a tightening surprise.

⁶Amberg *et al.* (2022) provide evidence on income showing a U-shaped effect, suggesting that quantile crossing may not be a major concern in our analysis.

5 Conclusion

In this study, we examined the influence of monetary policy on wealth inequality in the United States from 1989 to 2023 using a local projection approach. Our findings suggest that expansionary monetary policy shocks lead to greater wealth inequality, primarily by increasing the relative wealth of the highest income groups (the 99–100 and 80–99 income percentiles), while restrictive shocks tend to reduce inequality. However, this does not imply that lower-income households experience absolute wealth losses during periods of expansionary monetary policy. Instead, wealthier households accumulate wealth at a disproportionately higher rate, thereby exacerbating inequality. This conclusion is further supported by the observation that the effects of monetary policy shocks appear to be more pronounced on the asset side of wealth. Moreover, our results have significant implications for the growing HANK literature and the dynamics of cyclical inequality and for monetary policy transmission in general.⁷

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⁷HANK models generally require countercyclical inequality to generate aggregate-demand amplification.

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Online Appendix - The Impact of Monetary Policy on Wealth Inequality

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This online appendix contains the following sections. First, Appendix A provides an overview of general facts and descriptive statistics related to wealth inequality and monetary shocks. Second, Appendix B presents detailed results by decomposing wealth changes across asset and liability classes to further investigate the transmission channels of monetary policy on wealth. Third, Appendix C discusses the lag selection process and provides robustness checks using different lag specifications. Finally, Appendices D and E present subsample results for the periods 1990–2019 and 1990–2009, respectively.

A Data - Additional Information

In this section, we provide an overview of general facts and descriptive statistics related to wealth inequality, as well as monetary shocks. We start with wealth evolution (and its components) over time across income percentiles. (Specifically, we analyze the relative share of wealth, assets, and liabilities for each income percentile in relation to overall wealth, assets, and liabilities.) Figure A1 provides time series data regarding relative wealth (net worth), asset, and liability shares from DFA data.¹ In other words, the figure presents data on relative wealth (net worth), assets, and liabilities, expressed as a percentage of total wealth, assets, and liabilities, respectively, across different income percentiles.

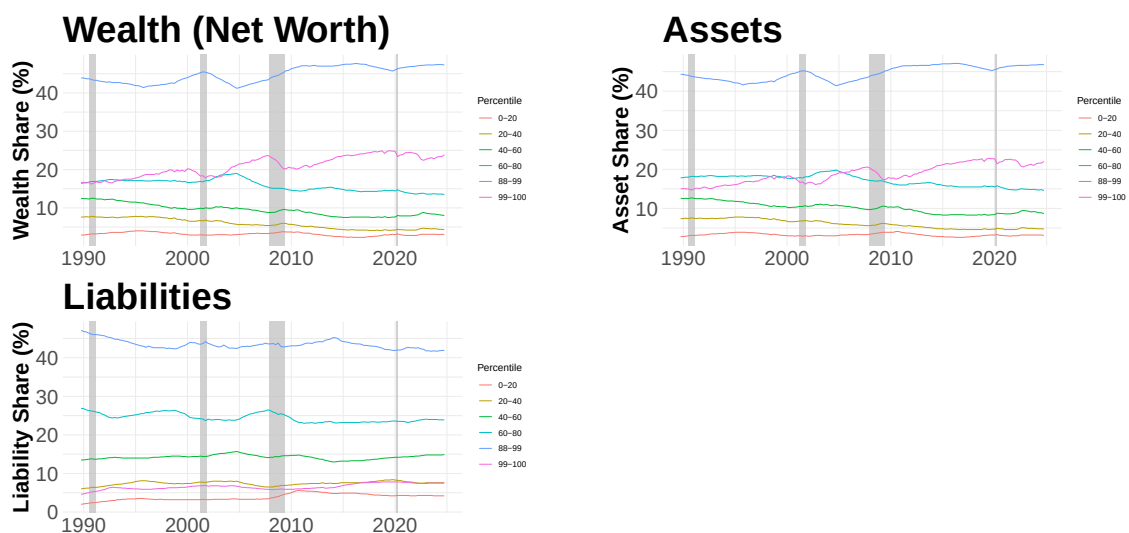
Figure A2 presents the distribution of wealth components across income percentiles, i.e., the distribution of relative shares of assets and liabilities for each income group. Mathematically, let A_p^j denote the holdings of asset class j by income group p . The boxplots are constructed using the ratio $A_p^j / \sum_j A_p^j$. For example, one of the boxplots illustrates what percentage of total assets are held in equities for the top 1% income group. This naturally applies also to other

¹As discussed in the main text, there are potential limitations with the DFA data, including the reliance on SCF-based interpolation and uncertainty regarding household mobility across wealth percentiles. However, the authors have implemented several adjustments to reconcile and improve the data. One key adjustment is the modification of the wealth estimates for the most wealthy to align with real-time estimates published by Forbes. This method enhances the credibility of our results for the top 1%. Additionally, we considered using the Real-time Inequality dataset by Blanchet et al. (2022), but this dataset does not have the same dimensions as the one used in the main analysis.

income percentiles. Similarly, we show the share of liabilities, such as mortgages or consumer credit, in total liabilities for each group over time. This shed some light on the transmission mechanisms of monetary policy shocks on wealth. For example, wealthier tend to hold a larger share of their assets in equities and private businesses, while lower-income groups allocate a greater proportion to real estate. Notably, real estate constitutes the dominant asset class across all income percentiles, except for the top 1% earners. On the liability side, wealthier households primarily carry mortgage debt rather than consumer credit. In the following part, we extend this analysis by decomposing wealth changes by asset and liability class to further explore these transmission channels.

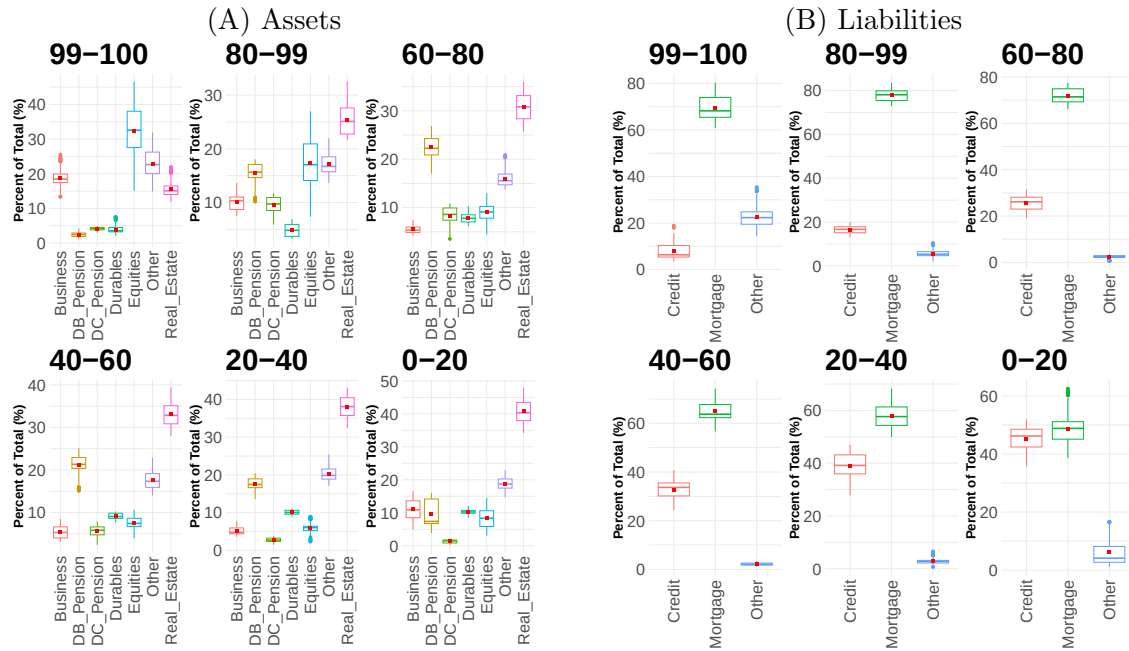
Next, we present the time series of monetary policy shocks and orthogonalized monetary policy shocks in Figure A3. These series capture unexpected changes in interest rates around FOMC announcements, with the raw monetary policy surprise reflecting observed rate changes and the orthogonalized series (used in the analysis) isolating the component unexplained by publicly available information prior to the announcements.

Figure A1: Wealth Evolution (and Its Components) Over Time



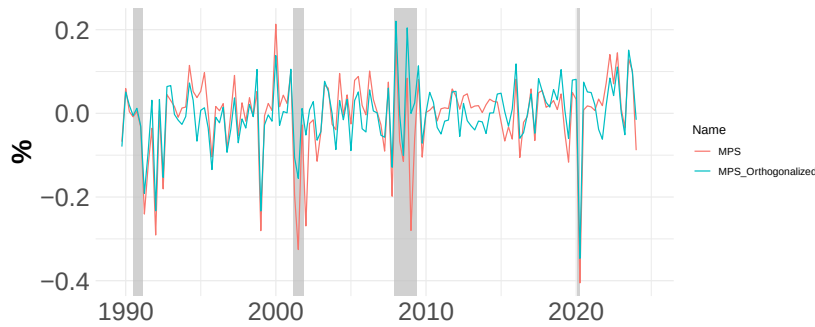
Note: The figures in the panel present the time series of DFA data, illustrating the distribution of wealth components across income percentiles. The top-left panel depicts relative wealth (net worth) percentage shares, while the top-right panel shows relative asset shares, and the bottom-left panel displays relative liability shares. The sample period spans from 1989:Q3 to 2023:Q4. Shaded regions denote NBER-dated recessions.

Figure A2: Boxplots of Main Components of Assets and Liabilities



Note: The figures in the panel present boxplots of DFA data, illustrating the distribution of wealth components across income percentiles. Red dots denote the mean. The left panel depicts asset shares, while the right panel shows liability shares. These panels display the distribution of relative shares of assets and liabilities for each income group. Business denotes private business. Business denotes private business, Equities denote corporate equities and mutual fund shares, DB.Pension denotes defined benefit pension entitlements, and DC.Pension denotes defined contribution pension entitlements. Credit denotes consumer credit.

Figure A3: Monetary Policy Shocks (MPS)



Note: The figure presents 30-minute changes in interest rates around FOMC announcements, using futures rates covering a four-quarter horizon following the announcement. Surprises are measured in percentage points. The red line (MPS) represents the raw monetary policy surprise based on observed rate changes, while the blue line (MPS Orthogonalized) depicts the surprise after being orthogonalized with respect to publicly available data preceding each FOMC announcement. The sample period spans from 1989:Q3 to 2023:Q4. Shaded regions denote NBER-dated recessions.

B Wealth Changes - Detailed Analysis

In this section, we present detailed results by decomposing wealth changes across asset and liability classes to further investigate the transmission channels of monetary policy on wealth. Using the DFA data's asset class classification, we first analyze the relative shares of asset and liability classes across income percentiles. Mathematically, let A_p^j denote the holdings of asset class j by income group p . The series are constructed using the ratio $A_p^j / \sum_p A_p^j$ for each j and p . For example, we examine the proportion of real estate holdings on overall real estate for which top earners account. This approach allows us to assess how the composition of household balance sheets evolves in response to monetary policy shocks. Given a large number of asset and liability categories, we report only the unrestricted point estimates without imposing parameter restrictions or displaying confidence intervals. Additionally, for comparison, we provide results on overall asset and liability shares, which remain consistent with those presented in the main text.

Our findings indicate that there is no single dominant driver of wealth changes; instead, multiple factors contribute (see Figure B1 for asset shares and Figure B2 for liability shares). For instance, the top 1% earners experience the most significant declines in private business holdings following a monetary policy shock, though equities also play a significant role. For the 80–99th income percentile, the observed effects are primarily driven by equities and real estate, whereas for the 60–80th percentile, equities appear to be the main driver. For lower-income groups, the results are more mixed, but generally, their relative share across most asset classes tends to increase. However, it is essential to note that our analysis focuses on the relative share of each income group within a specific asset or liability class. For example, since private businesses represent a relatively smaller asset class than equities, their overall impact on overall wealth is more limited. On the liability side, the most significant changes occur in other liabilities, but their overall effect remains limited due to their smaller size compared to mortgages and consumer credit.

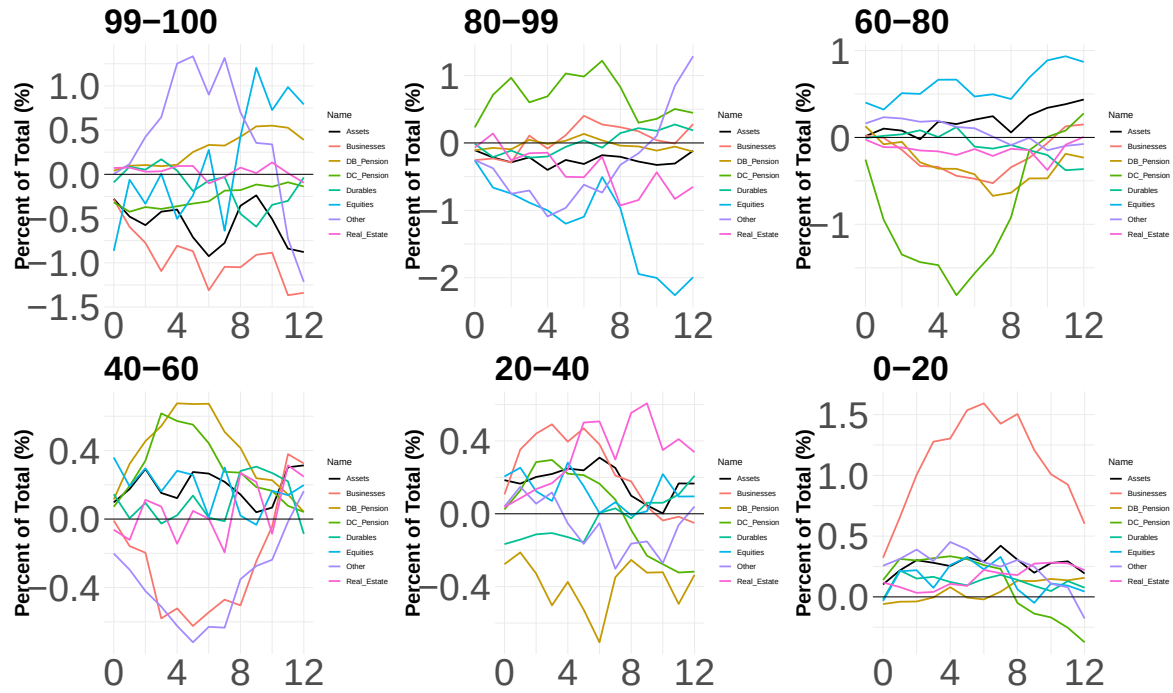
Furthermore, we also present results for percentage changes in wealth, expressed in real terms rather than as relative shares (see Figure B3 for assets and Figure B4 for liabilities). Specifically, we deflate assets and liabilities for each income group and asset/liability class using the Consumer Price Index (CPI) to adjust for inflation. (The variables are expressed in log form.) These results capture changes in real wealth rather than wealth inequality. Overall, the findings align with most of the literature, indicating that asset values decline in real terms following a monetary policy shock, with equities experiencing the most significant decline.²³

Finally, it is also crucial to recognize that our dataset captures final wealth, meaning that observed increases in wealth may stem from labor income rather than asset appreciation or depreciation. Unfortunately, we are unable to distinguish between these effects in our analysis.

²Regarding asset price appreciation and depreciation effects, monetary policy shocks influence the longer end of the yield curve only indirectly, making the results for fixed-income assets non-trivial. Additionally, for dividend-yielding assets, the standard capitalization formula $P = D/(r - g)$ applies; however, growth (g) itself may also be affected by monetary policy. Therefore, we interpret these effects with caution.

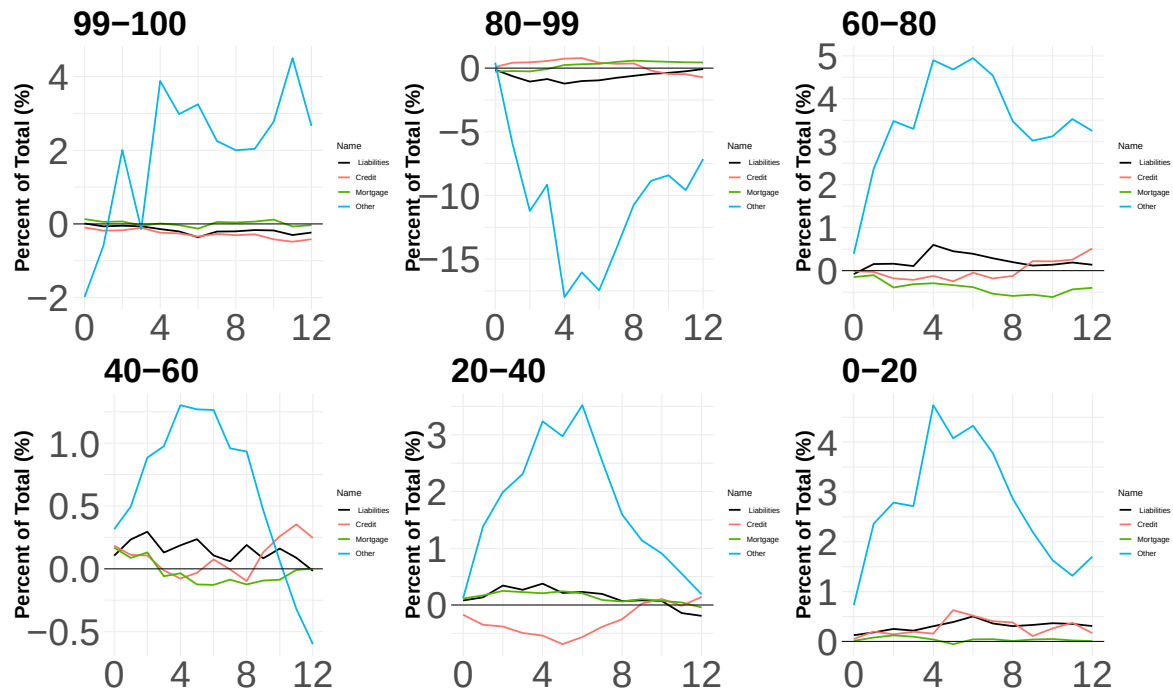
³Liabilities generally increase for low-income groups and decrease for high earners.

Figure B1: LP Results - Assets (Relative Shares)



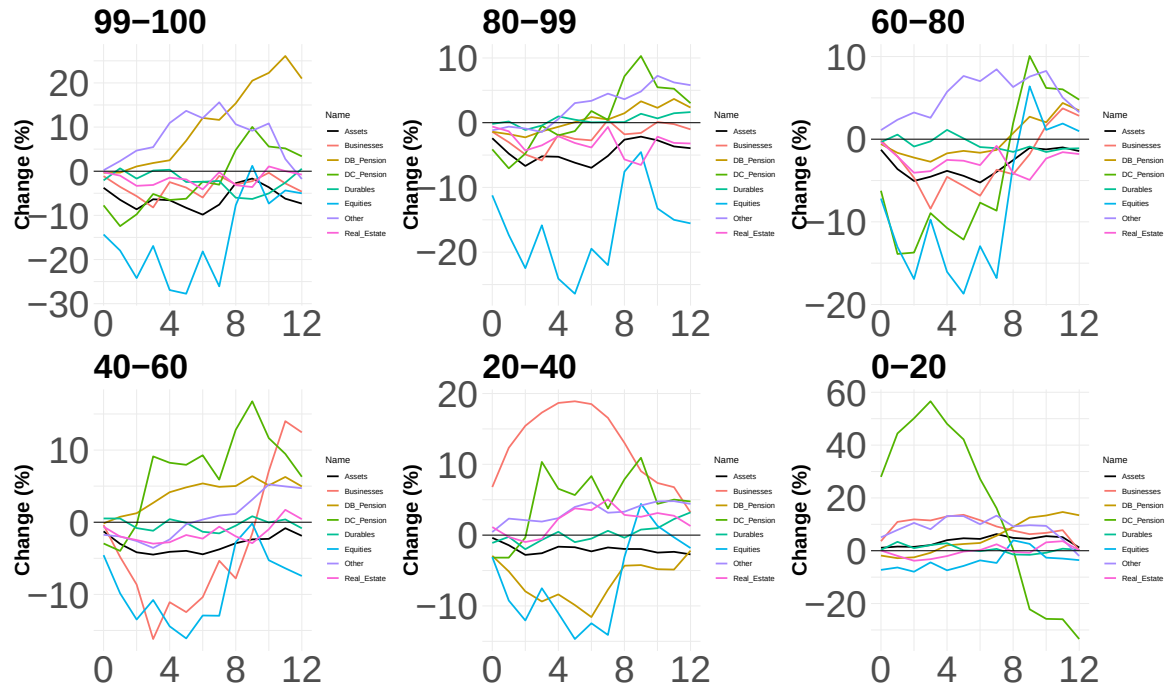
Note: The figures in the panel display the results for the relative share of each asset class across individual income percentiles (indicated by the title in each figure). We show unrestricted point estimates. The sample period spans from 1990:Q1 to 2009:Q4. The number of lags for control variables is set to 1. Business denotes private business, Equities denote corporate equities and mutual fund shares, DB_Pension denotes defined benefit pension entitlements, and DC_Pension denotes defined contribution pension entitlements.

Figure B2: LP Results - Liabilities (Relative Shares)



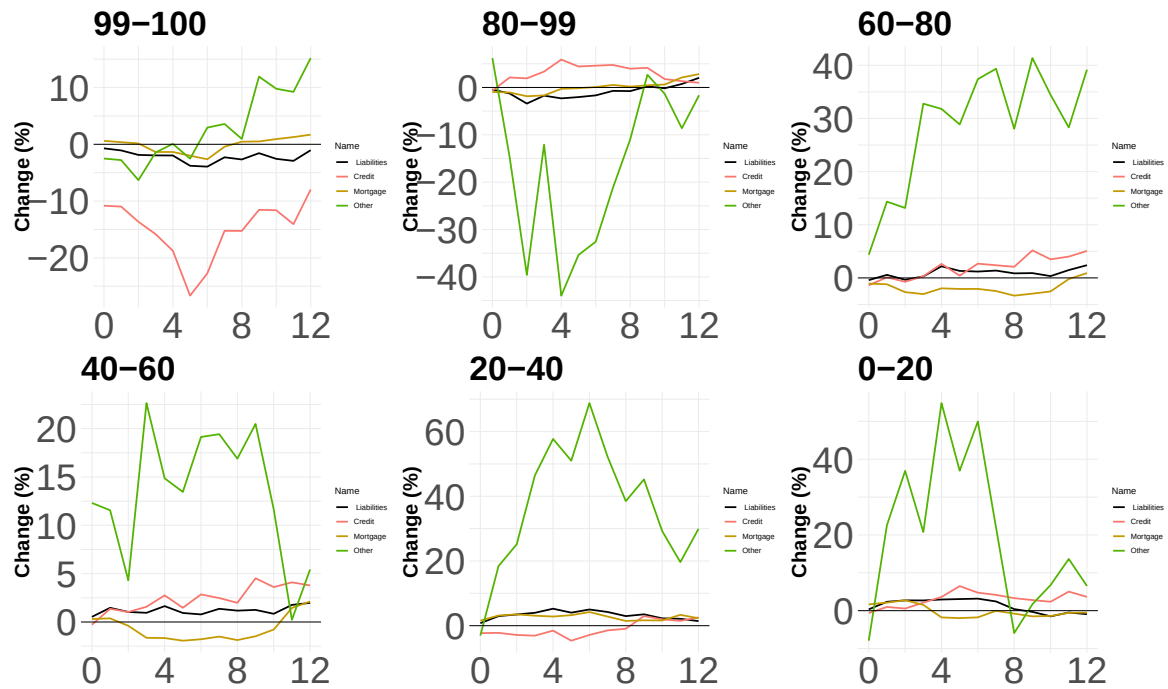
Note: The figures in the panel display the results for the relative share of each liability class across individual income percentiles (indicated by the title in each figure). We show unrestricted point estimates. The sample period spans from 1990:Q1 to 2009:Q4. The number of lags for control variables is set to 1. Credit denotes consumer credit.

Figure B3: LP Results - Assets (Real Values in Levels)



Note: The figures in the panel display the results for the percentage change in the real value of asset classes across individual income percentiles (indicated by the title in each figure). We show unrestricted point estimates. The sample period spans from 1990:Q1 to 2009:Q4. The number of lags for control variables is set to 1. Business denotes private business, Equities denote corporate equities and mutual fund shares, DB_Pension denotes defined benefit pension entitlements, and DC_Pension denotes defined contribution pension entitlements.

Figure B4: LP Results - Liabilities (Real Values in Levels)



Note: The figures in the panel display the results for the percentage change in the real value of liability classes across individual income percentiles (indicated by the title in each figure). We show unrestricted point estimates. The sample period spans from 1990:Q1 to 2009:Q4. The number of lags for control variables is set to 1. Credit denotes consumer credit.

C Robustness Check 1 (Different Lag Specification)

In this part, we discuss the lag selection process regarding the main results. The optimal lag structure was determined using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Figure C1 presents the corresponding information criteria values, showing that BIC favors a one-lag specification, while AIC prefers a two-lag specification. We report results based on a specification with two lags, while the main text provides estimates using a one-lag specification. The findings indicate that the results remain largely unchanged across different lag specifications. For reference, Figure C2 presents results for wealth (net worth), Figure C3 for assets, and Figure C4 for liabilities.

Figure C1: Information criteria

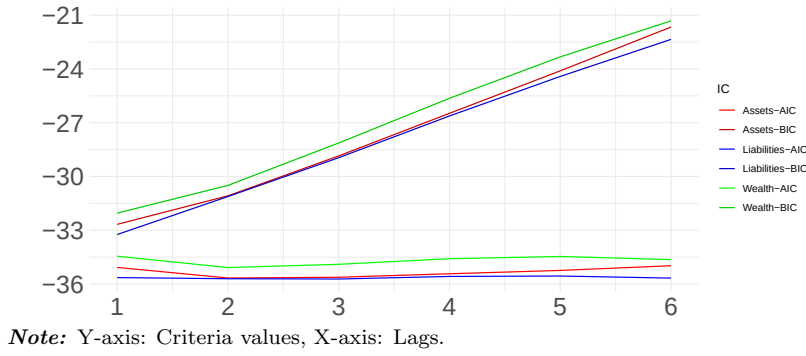
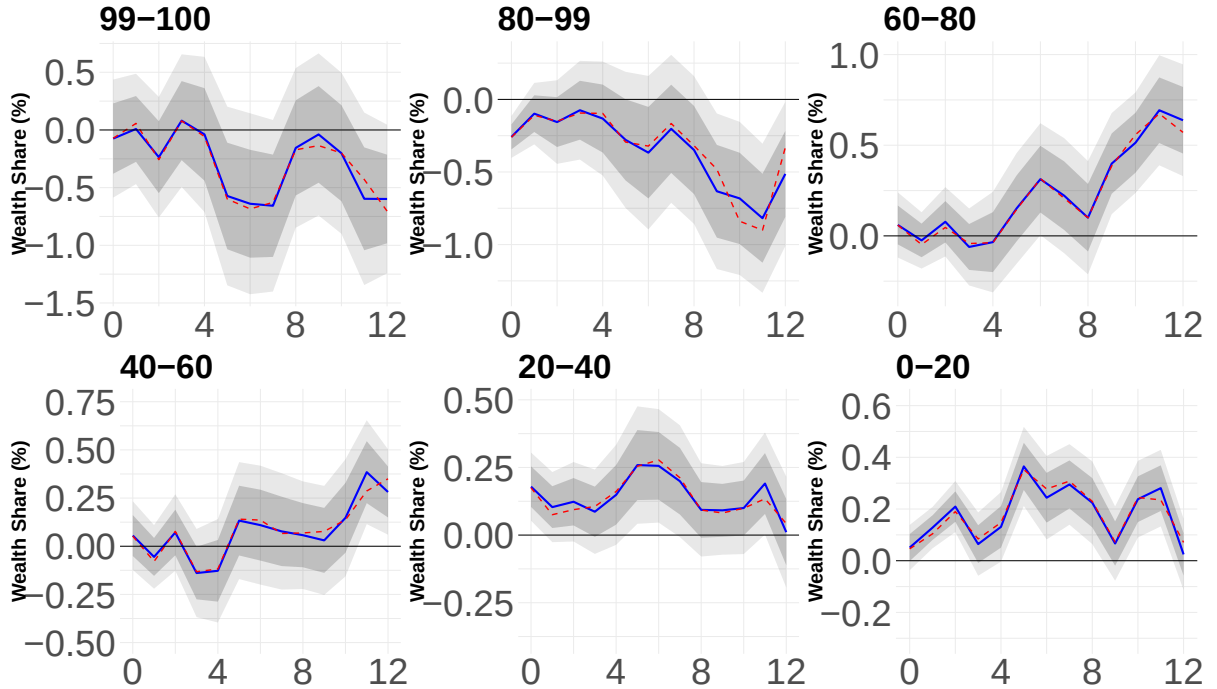
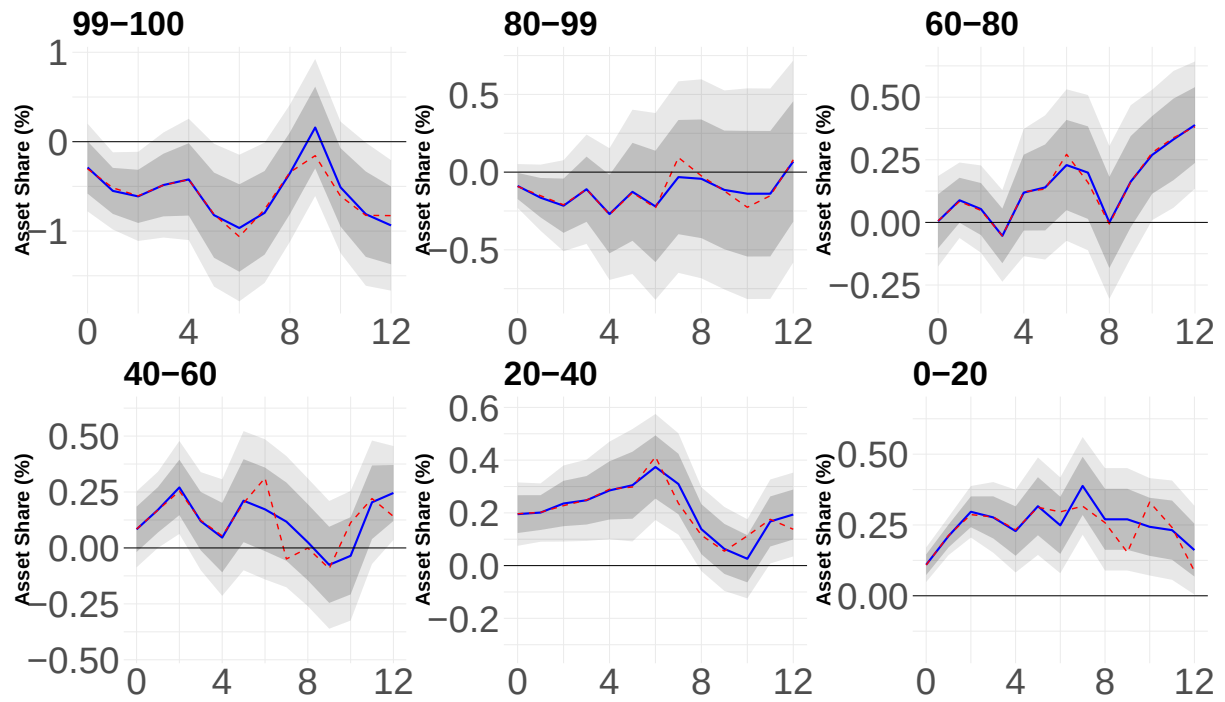


Figure C2: LP Results - Wealth (Net Worth)



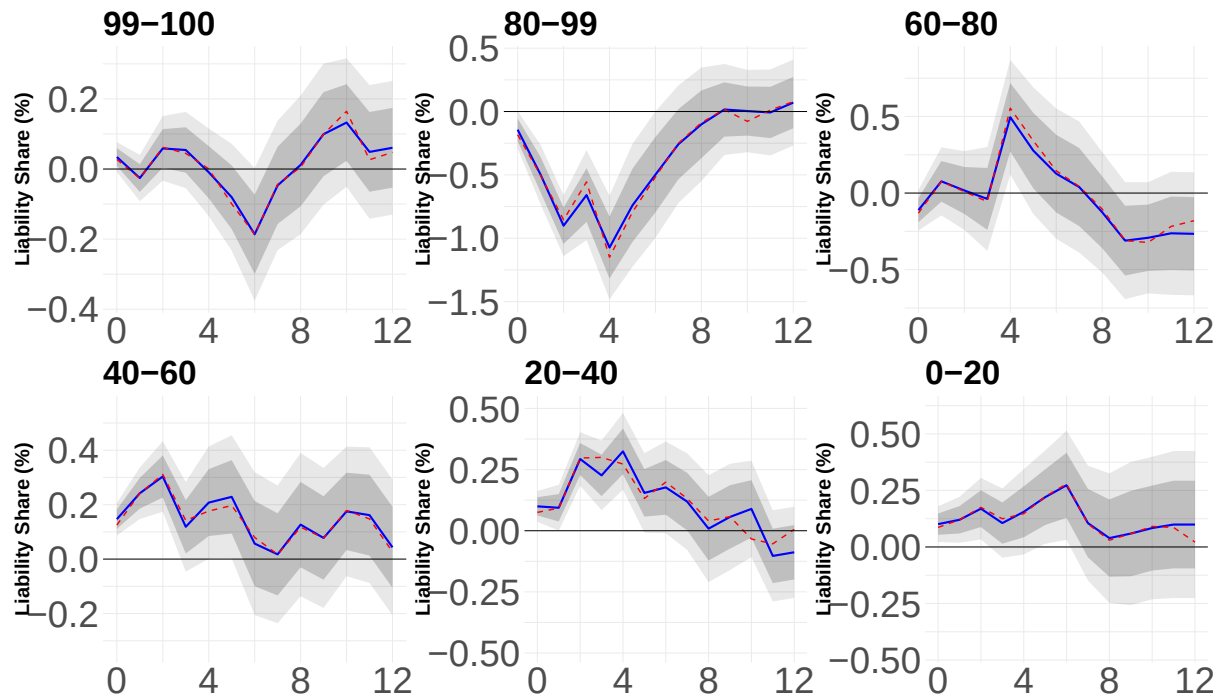
Note: The figures in the panel display the results for relative wealth share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2023:Q4. The number of lags for control variables is set to 2.

Figure C3: LP Results - Assets



Note: The figures in the panel display the results for relative asset share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2023:Q4. The number of lags for control variables is set to 2.

Figure C4: LP Results - Liabilities

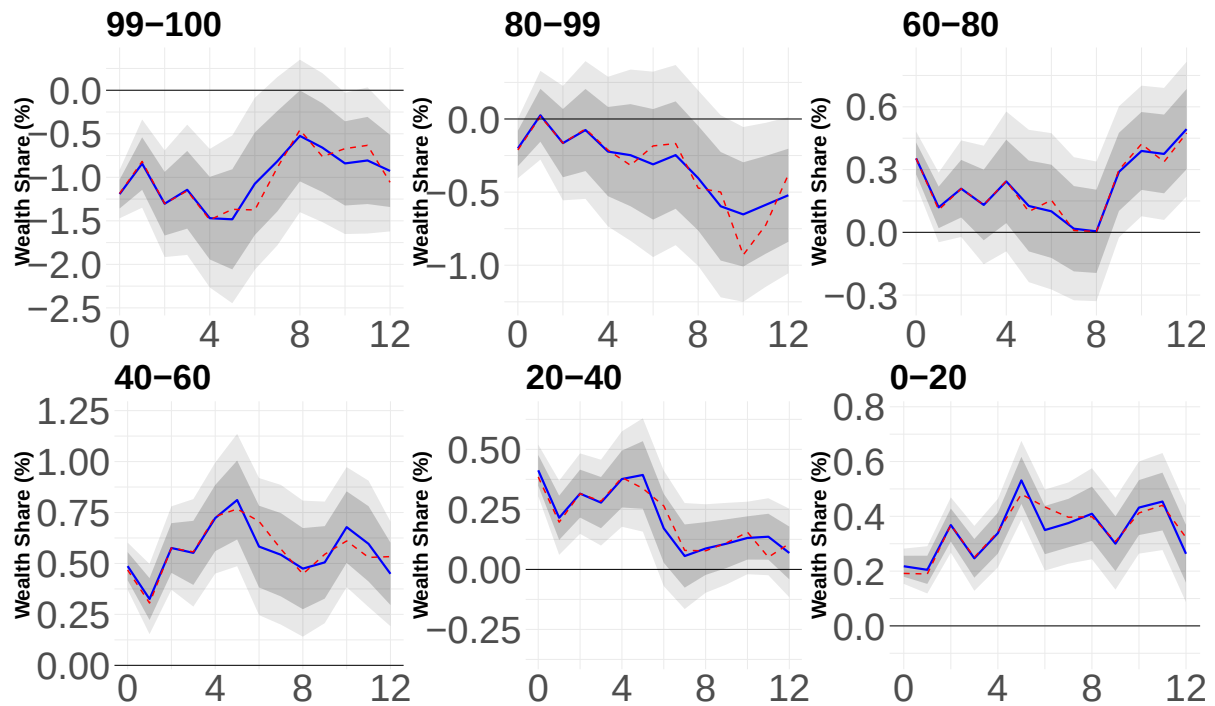


Note: The figures in the panel display the results for relative liability share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2023:Q4. The number of lags for control variables is set to 2.

D Robustness Check 2 (Subsample Results: 1990-2019)

In this section, we present subsample results for the period 1990-2019 for the relative wealth, asset or liability shares across individual income percentiles. Given the unprecedented nature of the COVID-19 pandemic, we re-estimate the model using only the pre-pandemic sample to assess the robustness of our findings.⁴ The results remain consistent and, if anything, appear more statistically significant. Nevertheless, our estimates still exhibit a relatively high degree of uncertainty, with statistical significance fluctuating between the 90% and 68% confidence levels. For reference, Figure D1 presents results for wealth (net worth), Figure D2 for assets, and Figure D3 for liabilities.

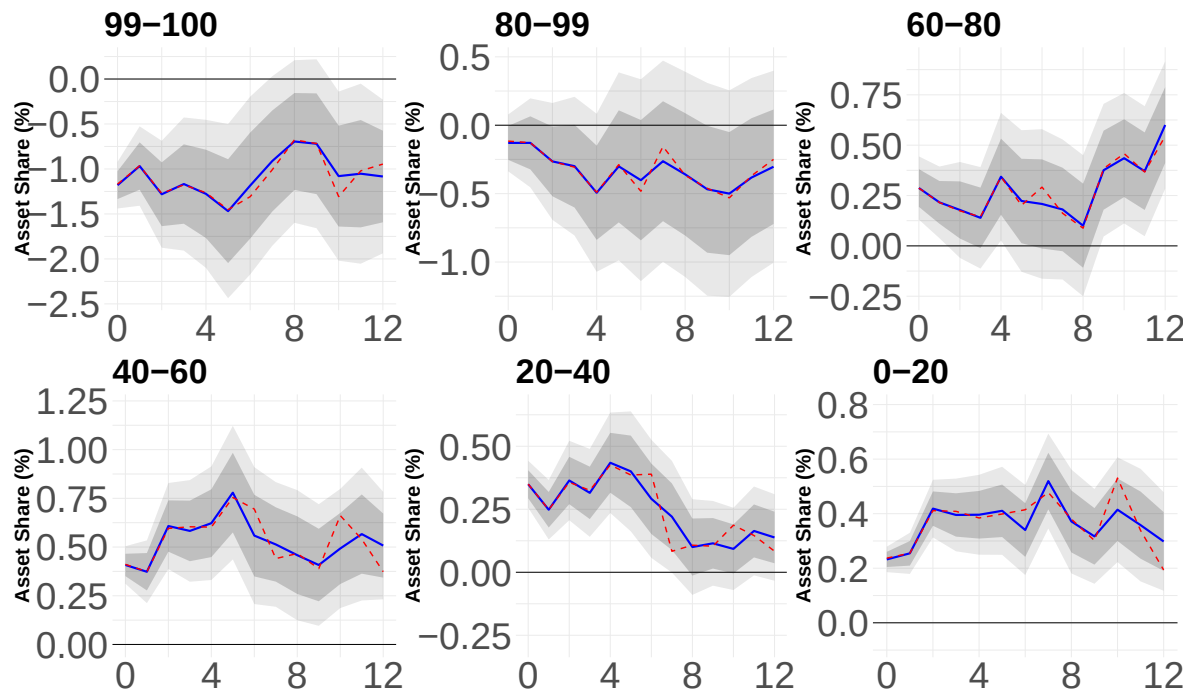
Figure D1: LP Results - Wealth (Net Worth)



Note: The figures in the panel display the results for relative wealth share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2019:Q4. The number of lags for control variables is set to 1.

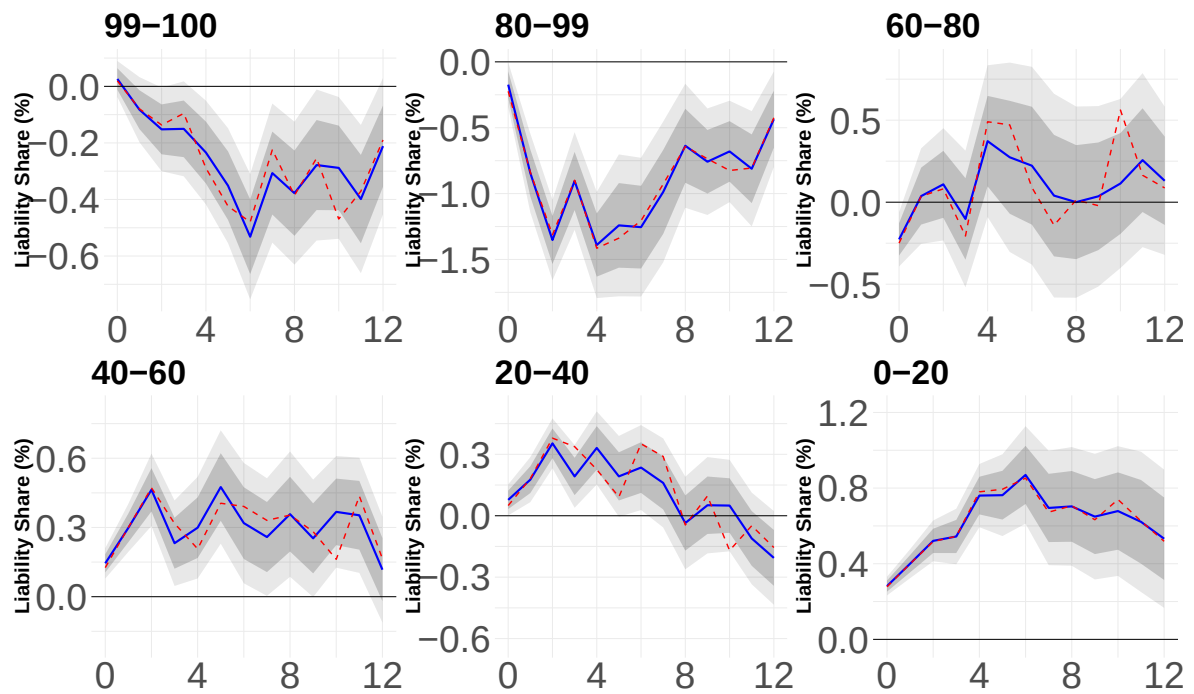
⁴While HAC standard errors account for heteroscedasticity, the COVID-19 pandemic may represent an outlier rather than a standard case of volatility.

Figure D2: LP Results - Assets



Note: The figures in the panel display the results for relative asset share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2019:Q4. The number of lags for control variables is set to 1.

Figure D3: LP Results - Liabilities

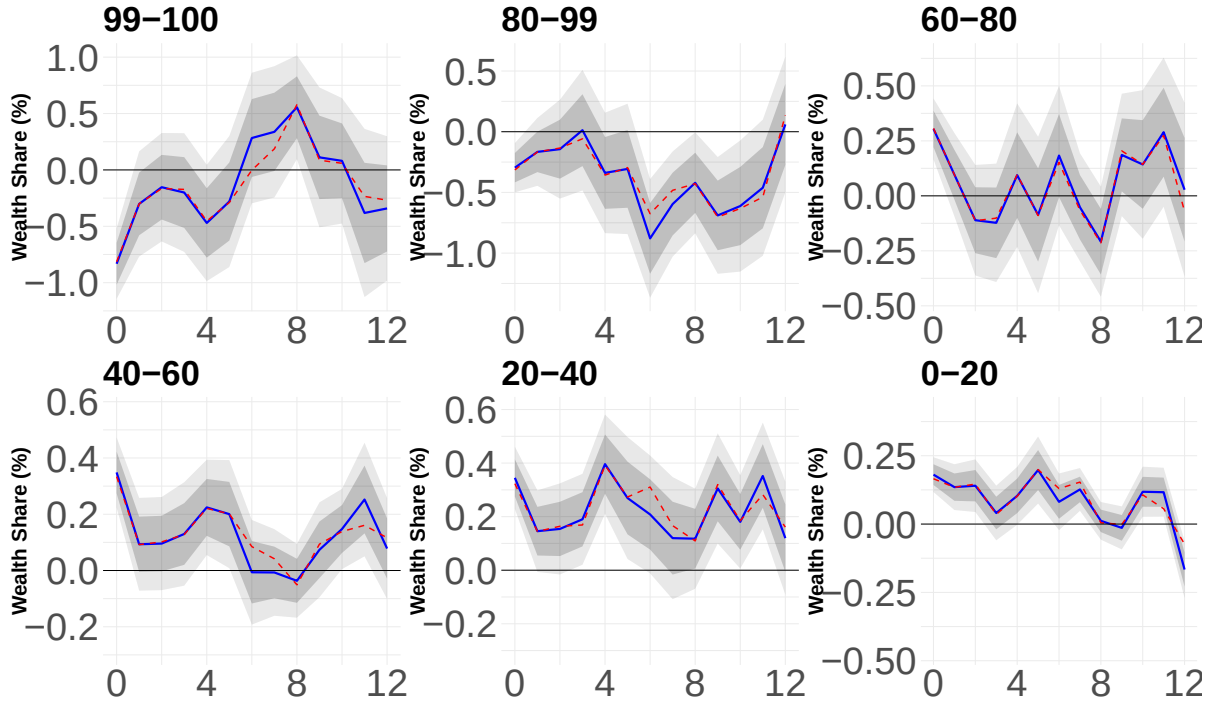


Note: The figures in the panel display the results for relative liability share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2019:Q4. The number of lags for control variables is set to 1.

E Robustness Check 3 (Subsample Results: 1990-2009)

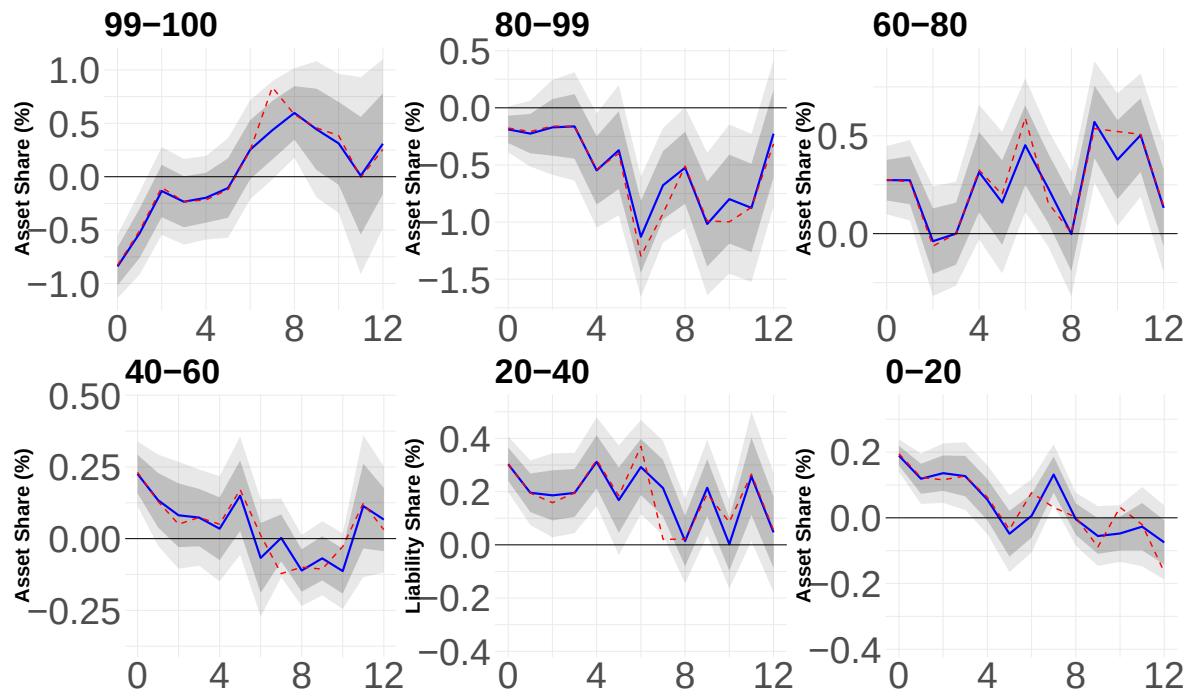
In this section, we present subsample results for the period 1990-2009 for the relative wealth/asset/liability shares across individual income percentiles. We re-estimate the model using only data up to 2010 to assess the robustness of our findings. The results largely remain consistent with the narrative that expansionary monetary policy shocks contribute to greater wealth inequality. However, the findings for the top 1% are more mixed, showing less clarity in the impact for this group. This may suggest the influence of zero lower bound period and quantitative easing, which may disproportionately inflate asset prices and benefit top earners. Additionally, our estimates exhibit a relatively high degree of uncertainty, with statistical significance primarily observed at the 68% confidence level. Figure E1 presents results for wealth (net worth), Figure E2 for assets, and Figure E3 for liabilities.

Figure E1: LP Results - Wealth (Net Worth)



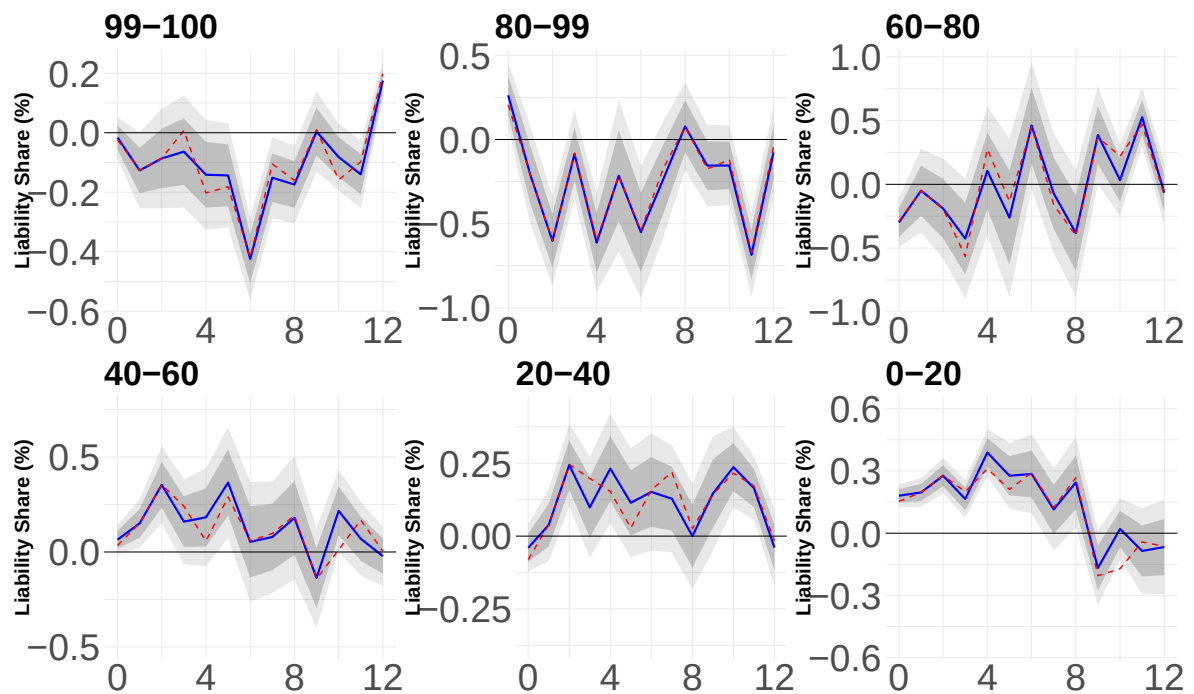
Note: The figures in the panel display the results for relative wealth share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2009:Q4. The number of lags for control variables is set to 1.

Figure E2: LP Results - Assets



Note: The figures in the panel display the results for relative asset share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2009:Q4. The number of lags for control variables is set to 1.

Figure E3: LP Results - Liabilities



Note: The figures in the panel display the results for relative liability share across individual income percentiles (indicated by the title in each figure). The blue line indicates an unrestricted point estimate. The red dashed line indicates a restricted point estimate. Shaded areas indicate 90% (light) and 68% (dark) confidence intervals for unrestricted estimation. The sample period spans from 1990:Q1 to 2009:Q4. The number of lags for control variables is set to 1.